

Abish Bhusal
Backend Software Engineer (Go,Python)
Bagmati, Kathmandu, Nepal
abish.bhusal@gmail.com

~ github.com/abishz17
[?] linkedin.com/in/ankitpaudel20
[?] abishbhusal.com.np

WORK EXPERIENCE

• Rara Labs Private Limited

Onsite / Lalitpur, Nepal

Backend Developer

Apr 2023 – Present

- **Backend Development:** - Contributed to the architecture, development, and database management of a highly available, multi-tenant, cloud-native backend for a comprehensive POS and business management SaaS platform.
- **System Architecture & Multi-Tenancy (Go):** - Engineered and maintained a layered backend in Go, enforcing strict separation of concerns to support a scalable, feature-rich system. Implemented and upheld the core multi-tenancy architecture (Shared Database), using request-scoped context to propagate the `organization_id` and ensure absolute data isolation in all database queries and business logic. Designed the asynchronous job processing system for handling resource-intensive background tasks like financial report generation and data synchronization, ensuring high API throughput.
- **Database Engineering & Administration (PostgreSQL):** - Authored and optimized complex SQL queries for the platform's core modules, including accounting, inventory, and reporting, leveraging advanced PostgreSQL features such as CTEs, window functions, and custom aggregates. Tuned performance by analyzing query plans (`EXPLAIN ANALYZE`) and adding indexes to significantly reduce API latency for heavy endpoints. Managed database connection pooling to reliably support thousands of concurrent tenant connections. Built and maintained a zero-downtime schema migration tool, enabling safe, version-controlled database updates. Designed and automated backup and point-in-time recovery strategies to ensure data integrity and disaster readiness.

SKILLS, LANGUAGES, INTERESTS

- **Languages:** English (Professional working proficiency), Nepali (Native proficiency)
- **Programming:** Golang, PostgreSQL, Python, C, Lua
- **Software Engineering:** Software Infrastructure, Database Administration, Concurrent Programming
- **Interests:** Video Games, Football, Reading superhero comics

EDUCATION

IOE, Pulchowk Campus , Tribhuvan University

Pulchowk, Lalitpur, Nepal

Bachelors in Computer Engineering

Aug 2018 – Jul 2023

- **Courses:** Operating Systems, Theory of Computation, Data Structures and Algorithms, C, OOP using C++, Artificial Intelligence and Machine Learning, Networking, Databases
- **Major Project:** Developed a Nepali OCR system using Convolutional Recurrent Neural Networks (CRNN) and Long Short-Term Memory (LSTM) networks, achieving 86% accuracy on a custom dataset of handwritten Nepali characters.
- **Extra Curricular:** Python instructor in a software fellowship organized by LOCUS , Organized a competitive programming hackathon called Yomari CodeCamp, Participated in multiple hackathons in my junior years

St. Xavier's College

Maitighar , Kathmandu, Nepal

High School major in Science

Aug 2016 – Jul 2018

- Graduated with Distinction, scoring 3.6/4.0 GPA in Science stream with Physics, Chemistry, and Maths.

CERTIFICATIONS

AWS Academy graduate-AWS Academy Cloud Foundations

AWS Academy

Issued by AWS Academy

Sep 2022

AWS Academy graduate-AWS Academy Cloud Architecting

AWS Academy

Issued by AWS Academy

Aug 2021

RESEARCH AND PUBLICATIONS

CRNN based Nepali OCR

Published on *Tribhuvan University*

Feb 2023